

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's stated design priorities — India-first sourcing, regional cultural content, sourcing from real exam papers **[Q11, Q12, Q24, Q25]** — are well aligned with the deployment's content needs. However, dataset analysis reveals two critical issues for the Hindi partition: (1) all 26 Hindi validation examples carry `is_translated=True` **[DATASET-D13]**, suggesting the Hindi partition may contain a much higher translated proportion than the documented overall ~25% **[Q47]**; (2) cross-language item duplication is pervasive **[D14, D15, D29]**, reducing effective Hindi-specific content. North India sub-regional content is sparse in the observed Hindi sample (only **DATASET-D3** on UP minerals). Current Affairs items are time-bounded to 2014–2022 events **[DATASET-D4, D25, D33]**, creating staleness risk for 2025–26 exam cycles. While translated items use fluent Devanagari with low code-mixing **[DATASET-D12, D21]**, the linguistic register may not match authentic Hindi-medium exam papers.

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q25	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D13	All 26 Hindi examples: <code>is_translated = True</code> — Every Hindi validation example in the sample is machine-translated from English
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
D4	मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कतिने स्वर्ण पदक जीते? ... आठ — Current Affairs question (2022) about archery medal count — time-bounded, may be stale
D12	रेडक्लिफ रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ वभाजित करती है? ... पाकस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूरगिर करने वाले वदिशी मुद्रा भंडार की हानिका तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but 'is_translated: True', confirms translation pipeline
D29	ഭരണഘടനാ ഭേദഗതി നിയമം വഴി ... 44-ാമത് ഭരണഘടനാ ഭേദഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication
D25	ਫਰਵਰੀ - ਮਾਰਚ 2022 ਵੱਚ ਹੋਈ ਰਾਜ ਵਧਿਨਸਭਾ ਚੋਣਾਂ ਵੱਚ, ਆਮ ਆਦਮੀ ਪਾਰਟੀ (AAP) ਨੇ ... ਪੰਜਾਬ — Current Affairs (2022 state elections) in Punjabi — politically specific, time-bounded
D33	ফব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বধিনসভা নব্বিবাচনে, আম আদমি পার্টি (AAP) ... পাঞ্জাব — 2022 state elections Current Affairs — AAP, UP, Goa all named; time-bounded content
D21	एक भौगोलिक क्शेत्त्र में नमिनलखिति वशिषिट वशिषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing

Strengths

- Source content is real Indian competitive exam papers [Q24, Q25], grounding the input in authentic exam ecosystems.
- India-first design explicitly includes culturally relevant subjects — local history, arts, festivals, laws [Q12, Q2].
- Constitutional law, Mughal/medieval North Indian history, and Indian GK are present in Hindi [DATASET-D1, D2, D3, D11, D18] — directly relevant to UPSC/SSC content.
- Observed Hindi items use clean Devanagari with minimal English code-mixing [DATASET-D12, D21], respecting the deployment's ~10% ceiling.

ID	Evidence
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q25	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q2	"With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science." (p.1)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D2	मुगल ग्रंथ 'मासि-ए-आलमगीरी' के रचयिता कौन हैं? ... साकी मुस्तईद खान — Mughal-period history question in Hindi — directly relevant to North India medieval history
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
D11	'मानसून' शब्द की उत्पत्तिकिस भाषा से हुई है? ... अरबी भाषा — General Knowledge question in Hindi (monsoon etymology) — accessible, pan-India GK
D18	नमिनलखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार थीं, जिन्हें कभी-कभी भारत की फ्रीदा काहलो के नाम से जाना जाता है? ... अमृता शेर्गलि — Arts & Culture GK in Hindi — pan-India Indian art figure
D12	रेडक्लिफ रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ विभाजित करती है? ... पाकिस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D21	एक भौगोलिक क्षेत्र में नमिनलखिति विशेषित विशेषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes, deployment requires both pan-India GK and North India sub-regional knowledge [elicitation Q2]. MILU partially addresses this — pan-India GK is well represented [DATASET-D1, D11, D18, D22], but North India sub-regional content (Chhath Puja, zamindari, North Indian freedom fighters) is sparse in observed Hindi sample (only DATASET-D3).

IC-2: Cultural alignment is partially achieved through India-first sourcing [Q11, Q24], but the dominant translation path for the Hindi partition (DATASET-D13: 100% is translated in sample) means linguistic-cultural register may not match Hindi-medium

exam conventions.

IC-3: No Western-specific content was observed; the benchmark is India-centric. However, translated engineering items [DATASET-D5, D6, D7, D36] inject construct-irrelevant content that mismatches central-exam syllabus.

IC-4: INSUFFICIENT DOCUMENTATION — paper does not report whether annotators or reviewers were Hindi-medium graduates or domain experts familiar with central-exam content [Q87 mentions only 'AI4Bharat volunteers'].

IC-5: Critical content issues: (a) probable over-representation of translated items in Hindi partition; (b) Current Affairs staleness (events 2014–2022 in a benchmark to be used for 2025–26 prep); (c) cross-language duplication [D14, D15, D29] reducing genuine Hindi-source diversity; (d) sparse North India sub-regional items observed.

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Q2	"With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science." (p.1)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
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D18	नमिनलखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार थी, जिन्हें कभी-कभी भारत की फ़र्दा काहलो के नाम से जाना जाता है? ... अमृता शेर्गलि — Arts & Culture GK in Hindi — pan-India Indian art figure
D22	What was the first capital of the Bahamani Kingdom? ... Gulbarga — India-specific historical knowledge (Deccan Sultanate) — same question appears across all languages
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English
D5	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... पल्सेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep
D36	एएम की बैंडविड्थ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, `is_translated: True`; unlikely in UPSC GS
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
Q44	"For subjects with insufficient questions, we sampled questions from the English set from that subject and translated them into the required language using GPT-4O." (p.4)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D29	ഭരണഘടനാ ഭേദഗതി നിയമം വഴി ... 44-ാമത് ഭരണഘടനാ ഭേദഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication

Information Gaps

- True translation rate in the Hindi test split (vs. observed 100% in 26-item validation sample) is unknown.
- Proportion of MILU Hindi items aligned with central exams vs. state PSCs is undocumented.
- Item-level North India sub-regional content breakdown is not published.
- Annotator demographics — whether reviewers were Hindi-medium graduates familiar with the target student linguistic register — not reported.

Requires Expert Verification

- A Hindi-medium UPSC/SSC subject-matter expert should review a stratified sample of MILU Hindi test items for: (a) authenticity of register vs. authentic Hindi-medium exam papers, (b) translation artifact rate, (c) syllabus alignment with central exams, (d) North India sub-regional content presence.

Remediation

Gap 1: Hindi partition appears to contain a high proportion of translated items (DATASET-D13: 100% in observed sample), pervasive cross-language duplication, and stale Current Affairs.

Recommendation: Curate a deployment-specific Hindi subset: (a) prefer items with is_translated=False where they exist; (b) deduplicate across languages; (c) replace pre-2023 Current Affairs items with a fresh annual current-affairs probe; (d) ensure North India sub-regional items (Chhath Puja, zamindari, North Indian freedom fighters) are sufficiently represented via custom item authoring.

Gap 2: North India sub-regional content (Chhath Puja, zamindari, regional freedom fighters) is sparsely observed in the Hindi sample (1 of 26 items).

Recommendation: Author or source ~200–500 additional Hindi items specifically covering North India sub-regional content within central-exam framing, drawing from authentic UPSC/SSC previous-year papers.

ID	Evidence
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English